Lakshya JEE 2.0 2025

Maths

Matrices

DPP: 2

Q1 If $A=\begin{bmatrix}2 & -3\\ -4 & 1\end{bmatrix}$, then adj $\left(3A^2+12A\right)$ is equal to:

(A) $\begin{bmatrix}72 & -63\\ -84 & 51\end{bmatrix}$

(B) $\begin{bmatrix}72 & -84\\ -63 & 51\end{bmatrix}$

(C) $\begin{bmatrix}51 & 63\\ 84 & 72\end{bmatrix}$

(D) $\begin{bmatrix}51 & 84\\ 63 & 72\end{bmatrix}$

Q2 If $A(\alpha)=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$, then the matrix of $A^2(\alpha)$ is equal to:

(A) $A(2\alpha)$

(B) $A(\alpha)$

(C) $A(3\alpha)$

(D) $A(4\alpha)$

Q3 If $A=\begin{bmatrix}1 & 2 & x\\ 3 & -1 & 2\end{bmatrix}$ and $B=\begin{bmatrix}y\\ x\\ 1\end{bmatrix}$ be such that $AB=\begin{bmatrix}6\\ 8\end{bmatrix}$, then

(A) $y=2x$ (B) $y=-2x$

(C) $y=x$ (D) $y=-x$

Q4 If $A=\begin{bmatrix}a & b\\ b & a\end{bmatrix}$ and $A^2=\begin{bmatrix}\alpha & \beta\\ \beta & \alpha\end{bmatrix}$, then:

(A) $\alpha=2ab, \beta=a^2+b^2$

(B) $\alpha=a^2+b^2, \beta=ab$

(C) $\alpha=a^2+b^2, \beta=2ab$

(D) $\alpha=a^2+b^2, \beta=a^2-b^2$

Q5 If $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ a & 2 & b\end{bmatrix}$ is a matrix satisfying the equation $AA^T=9I$, where I is 3×3 identity matrix, then the ordered pair (a,b) is equal to:

(A) $(2,1)$ (B) $(-2,-1)$

(C) $(2,-1)$ (D) $(-2,1)$

Q6 If $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$, then for $n\in N$, A^n is equal to

(A) $\begin{bmatrix}1+2n & -4n\\ n & 1-2n\end{bmatrix}$

(B) $\begin{bmatrix}1-2n & 4n\\ n & 1+2n\end{bmatrix}$

(C) $\begin{bmatrix}1+2n & 4n\\ -n & 1+2n\end{bmatrix}$

(D) $\begin{bmatrix}1-2n & -4n\\ -n & -1+2n\end{bmatrix}$

Q7 If $A=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$, then A^{2008} is equal to

(A) $\begin{bmatrix}2008 & 0\\ 1 & 1\end{bmatrix}$

(B) $\begin{bmatrix}1 & 0\\ 2008 & 1\end{bmatrix}$

(C) $\begin{bmatrix}1 & 0\\ 1 & 2008\end{bmatrix}$

(D) $2007\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$

Q8 Minimum number of zeroes in diagonal matrix A having 25 elements is m and diagonal matrix B having 16 elements is n then value of $m-n$ is :

(A) 8 (B) 12
(C) 15 (D) 20

Q9 Which of the following is incorrect
(A) $\mathrm{Tr}(\mathrm{A}+\mathrm{B})=\mathrm{Tr}(\mathrm{A})+\mathrm{Tr}(\mathrm{B})$
(B) $\mathrm{Tr}(\mathrm{A}-\mathrm{B})=\mathrm{Tr}(\mathrm{A})-\mathrm{Tr}(\mathrm{B})$
(C) $\mathrm{Tr}(\mathrm{A})=n$, where A is scalar matrix
(D) $\mathrm{Tr}(k\,\mathrm{A})=k\,\mathrm{Tr}(\mathrm{A})$ where k is scalar

Q10 The value of x for which the matrix product $\begin{bmatrix}2 & 0 & 7\\ 0 & 1 & 0\\ 1 & -2 & 1\end{bmatrix}\begin{bmatrix}-x & 14x & 7x\\ 0 & 1 & 0\\ x & -4x & -2x\end{bmatrix}$ equals an identity matrix, is
(A) $1/2$ (B) $1/3$
(C) $1/4$ (D) $1/5$

Q11 If $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$, then the value of α for which $\mathrm{A}^2=\mathrm{B}$, is
(A) 1 (B) -1
(C) 4 (D) no real values

Q12 Let $P=\begin{bmatrix}1 & 0 & 0\\ 4 & 1 & 0\\ 16 & 4 & 1\end{bmatrix}$ and I be the identity matrix of order 3. If $\mathrm{Q}=[a_{ij}]$ is a matrix such that $\mathrm{P}^{50}-\mathrm{Q}=\mathrm{I}$, then $\frac{a_{31}+a_{32}}{a_{21}}$ equals
(A) 52 (B) 103
(C) 201 (D) 205

Q13 Let $\mathrm{A}=\begin{pmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{pmatrix}, (a\in\mathrm{R})$ such that $A^{32}=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$. Then a value of α is.
(A) $\pi/16$ (B) 0
(C) $\pi/32$ (D) $\pi/64$

Q14 Let α be a root of equation $x^2+x+1=0$ and the matrix $A=\frac{1}{\sqrt{3}}\begin{bmatrix}1 & 1 & 1\\ 1 & \alpha & \alpha^2\\ 1 & \alpha^2 & \alpha^4\end{bmatrix}$, is equal to
(A) A^3
(B) A^2
(C) I_3
(D) A

Q15 If $\mathrm{A}=\begin{bmatrix}-4 & -1\\ 3 & 1\end{bmatrix}$, then the determinant of the matrix $\left(A^{2016}-2A^{2015}-A^{2014}\right)$, is
(A) 2014 (B) 2016
(C) -175 (D) -25

Q16 If $\mathrm{P}=\begin{bmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix}, \mathrm{A}=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ and $\mathrm{Q}=\mathrm{PAP}^{\mathrm{T}}$, then $\mathrm{P}^{\mathrm{T}}\mathrm{Q}^{2015}\mathrm{P}$, is
(A) $\begin{bmatrix}2015 & 1\\ 0 & 2015\end{bmatrix}$
(B) $\begin{bmatrix}1 & 2015\\ 0 & 1\end{bmatrix}$
(C) $\begin{bmatrix}0 & 2015\\ 0 & 0\end{bmatrix}$
(D) $\begin{bmatrix}2015 & 0\\ 1 & 2015\end{bmatrix}$

Answer Key

Q1	(C)	Q9	(C)
Q2	(A)	Q10	(D)
Q3	(A)	Q11	(D)
Q4	(C)	Q12	(B)
Q5	(B)	Q13	(D)
Q6	(A)	Q14	(A)
Q7	(B)	Q15	(D)
Q8	(A)	Q16	(B)

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